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192572092

CSE-AI

PYTHON

CLASSWORK 4.8.26

Q1 Union of Sets

5 marks

Write a Python program to find the union of two sets.

Input:

A={1,2,3}

B={3,4,5}

ouput:

{1,2,3,4,5}

Your Code

```
1 A = {1, 2, 3}
2 B = {3, 4, 5}
3
4 C = A | B
5
6 print('{ ' + ', '.join(map(str, sorted(C))) + ' }')
```

Input

stdin input

Output

{1,2,3,4,5}

Q2 Intersection

5 marks

Write a Python program to find the intersection of two sets.

Input:

A={1,2,3}

B={2,3,4}

ouput:

{2,3}

Your Code

```
1 A = {1, 2, 3}
2 B = {2, 3, 4}
3
4 C = A.intersection(B)
5
6 print('{ ' + ', '.join(map(str, sorted(C))) + ' }')
```

Input

stdin input

Output

{2,3}

Q3 Difference

5 marks

Write a Python program to find the difference of two sets.

Input:

A={1,2,3}

B={2,3,4}

ouput:

{1}

Your Code

```
1 A = {1, 2, 3}
2 B = {2, 3, 4}
3
4 C = A.difference(B)
5
6 print('{ ' + ', '.join(map(str, sorted(C))) + ' }')
```

Input

stdin input

Output

{1}

Q4 Symmetric Difference

5 marks

Write a Python program to find the symmetric difference of two sets.

Input:

A={1,2,3}

B={2,3,4}

ouput:

{1,4}

Your Code

```
1 A = {1, 2, 3}
2 B = {2, 3, 4}
3
4 C = A.symmetric_difference(B)
5
6 print('{ ' + ', '.join(map(str, sorted(C))) + ' }')
```

Input

stdin input

Output

{1, 4}

Q5 Subset

5 marks

Write a Python program to check whether one set is a subset of another.

Input:

A={2,3}

B={1,2,3,4}

output:

Subset

Your Code

```
1 A = {2, 3}
2 B = {1, 2, 3, 4}
3
4 if A.issubset(B):
5     print('Subset')
6 else:
7     print('Not Subset')
```

Input

stdin input

Output

Subset

[< B](#)**Q6 Superset**

5 mark

Write a Python program to check whether one set is a superset of another.

Input:

A={1,2,3,4}

B={2,3}

output:

Superset

Your Code

```
1 A = {1, 2, 3, 4}
2 B = {2, 3}
3
4 if A.issuperset(B):
5     print('Superset')
6 else:
7     print('Not Superset')
```

Input

stdin input

Output

Superset

Q7 Disjoint Sets

5 marks

Write a Python program to check whether two sets are disjoint.

Input:

A={1,2}

B={3,4}

ouput:

Disjoint

Your Code

```
1 A = {1, 2}
2 B = {3, 4}
3
4 if A.isdisjoint(B):
5     print('Disjoint')
6 else:
7     print('Not Disjoint')
```

Input

stdin input

Output

Disjoint

Q8 Remove Duplicates

5 marks

Write a Python program to remove duplicate elements from a list using a set.

Input:

List: 1 2 2 3 3 4

ouput:

{1,2,3,4}

Your Code

```
1 lst = [1, 2, 2, 3, 3, 4]
2
3 s = set(lst)
4
5 print('{' + ','.join(map(str, sorted(s))) + '})')
```

Input

stdin input

Output

{1,2,3,4}

Visible Test Cases

Test Case 1



Write a Python program to find the common elements in three sets.

Input:

A={1,2,3}

B={2,3,4}

C={2,5}

Output:

{2}

Your Code

```
1 A = {1, 2, 3}
2 B = {2, 3, 4}
3 C = {2, 5}
4
5 D = A & B & C
6
7 print('{' + ','.join(map(str, sorted(D))) + '}' )
```

Input

stdin input

Output

{2}

Visible Test Cases

Test Case 1



Q10 Missing Numbers

5 marks

Write a Python program to find the missing numbers from 1 to 10 using sets.

Input:

Given={1,2,4,5,7,10}

Output:

{3,6,8,9}

Your Code

```
1 given = {1, 2, 4, 5, 7, 10}
2
3 all_numbers = set(range(1, 11))
4
5 missing = all_numbers - given
6
7 print('{' + ','.join(map(str, sorted(missing))) + '}' )
```

Input

stdin input

Output

{3,6,8,9}

Visible Test Cases

Test Case 1



5.00 / 5 marks